

The Hand That Gives

*Pre-PIE *de- and the polyfunctional architecture: the body-part noun *de-r-, the heteroclititic gift-noun *de- + *-u-ṛ-/-(e/o)n-, and the morphological skeleton of Papers 1–4*

Paper 1 of the *der-* project.

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Abstract

A recognized comparative cluster — Greek δῶρον “gift; palm-breadth,” Hesychian δάρις “span of all fingers,” Celtic **durno-* “fist, hand” (Old Irish *dorn*, Welsh *dwrn*, Breton *dorn/dourn*, Gaulish *Durnomagus*), and Latvian *dūfe* “fist” — has been treated as a puzzle since Pokorny (IEW 203). Schrijver (1995: 58) lists these forms as “possible cognates” and describes the formal relation between δωρ- and δαρ- as “obscure.” Matasović (2009, s.v. **durno-*) has concluded that no satisfactory Indo-European etymology exists and has classified the Celtic word as “probably a loanword from a non-IE language.”

This paper offers the missing etymology and, in the same movement, sets out the morphological architecture of the broader four-paper project. The proposal is that pre-PIE had a polyfunctional element **de-* “open hand, extended hand,” which underwent multiple parallel derivations. The lexical branch — Paper 1’s specific contribution — has two surface formations on the same root: the locative-extended body-part noun **de- + *-r-* (giving Hesychian δάρις, Old Irish *derna*, and Continental Celtic **DERNV*) and the heteroclititic gift-noun **de- + *-u-ṛ-/-(e/o)n-* (giving Greek δῶρον / δάνος, Latin *dōnum*, Sanskrit *dārú-*, Celtic **durno-*, and Latvian *dūfe*).

The relation between δωρ- and δαρ- is now substantially clarified by Clayton (2023), whose dissertation reconstructs the Indo-European **-wṛ-/-(e/o)n-* heteroclitite class and its widely-attested PNIE metathesis **wṛ > *ru*. Clayton independently identifies **deh₃-* as a root that forms heteroclitites of this class. Under his framework, Greek δῶρον is derivable as the metathesized strong stem; Celtic **durno-* is the same strong stem with subsequent revocalization (cf. PIIr. **druna-* > OKhotanese *durna-*; Bailey 1961). The Celtic *u* is therefore not problematic — as Beekes (EDG s.v. δῶρον) treated it — but inherited PNIE **-wṛ-* metathesized and revocalized, as Clayton’s framework predicts and as Lubotsky’s (1994) σύρξ-type mechanism phonologically motivates.

Beyond the lexical etymology, the paper sets out the project’s morphological skeleton (§2) and identifies which branches each subsequent paper develops. We assess the probability of the Paper 1 thesis at 30–40%. The marginal

probability for the framework as a whole is in the range 1–4%.

1. Introduction

A recognized comparative cluster has been without a satisfactory etymology for over a century. Pokorny’s *Indogermanisches etymologisches Wörterbuch* (IEW 203) grouped together Greek δῶρον “length of the hand,” Hesychian δάρις “span of all fingers,” Old Irish *dorn*, Welsh *dwrn*, Gaulish *Durnomagus*, and Latvian *dūfe* as members of a single comparative set. Schrijver (1995: 58) lists these forms as “possible cognates” and describes the formal relation between δωρ- and δαρ- as “obscure.” Matasović (2009, s.v. **durno-*) concludes that the IE etymology “cannot be accepted” and classifies the Celtic word as “probably a loanword from a non-IE language.” The comparative cluster is real; the etymology has been missing.

Three formal puzzles have driven the difficulty. First, the relation between Greek δῶρον (with ω) and Hesychian δάρις (with α) is clearly not accidental, but no standard reconstruction has explained the alternation within the established laryngeal framework to general satisfaction. Second, the consistent *u* of Celtic **durno-* and Gaulish *Durno-* has resisted systematic derivation from either a bare **der-* or the give-verb **deh₃-*. Third, the semantic gap between “hand, fist, palm” in Celtic and Baltic and “gift” in Greek and Latin has seemed too large for coincidence but too wide to bridge.

The present paper proposes a unified answer to all three puzzles, and in the same movement sets out the morphological architecture of the broader four-paper project. Pre-PIE had a polyfunctional element **de-* “open hand, extended hand,” productive across multiple derivational pathways. Two of those pathways produce the lexical branch the present paper documents. A locative extension **de-* + **-r-* produces the body-part noun **de-r-* (giving δάρις, *derna*, **DERNV*). A heteroclitic extension **de-* + **-u-ṛ-/w(e/o)n-* produces the gift-noun family (giving δῶρον / δάνος, Latin *dōnum*, Sanskrit *dārú-*, Celtic **durno-*, Latvian *dūfe*). The “obscurity” Schrijver flagged is the relation between two derivational pathways operating on the same bare element. The Celtic *u* is the inherited PNIE reflex of **-wṛ-* via the Clayton (2023) metathesis.

2. The polyfunctional architecture

2.1 Pre-PIE **de-* as a polyfunctional element

Paper 3 (Chavis & Claude 2026c) reconstructs **de-* as a polyfunctional pre-PIE morpheme with body-part, directional, and verbal functions. Body-part terms cross-linguistically grammaticalize into verbs (Heine & Kuteva 2002), into adpositions, and into spatial particles. “Hand” is among the most productive

sources of such grammaticalization: hand → give/take → benefactive marker; hand → side → contrastive marker.

The directional particle function is independently attested as PIE **de/do* “to, toward” — preserved as the Greek allative clitic *-δε* (productive in Mycenaean Greek by 1400–1200 BCE), Old Church Slavonic *do*, English *to* (Proto-Germanic **tō*), and Celtiberian *DV* in the Bronze of Novallas (Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45). Paper 3 develops this branch in detail. The verbal function is independently attested as the give-verb **deh₃-* (Sanskrit *dádāti*, Greek *δίδωμι*, Latin *dō/dare*), which Paper 3 §5 decomposes as bare **de-* plus a verbal extension **-h₃-*, and as Hittite *dā-* (Paper 4).

2.2 The skeleton: derivational pathways on the same root

The architecture across the four-paper framework is shown in the table below. Each row is a different derivational outcome on the same pre-PIE root; they are aspects of a single morphological system. Bauhaus (2019) reconstructs an analogous paradigmatic opposition between locative **-r* and directive **-o* on a class of nominal roots, providing published external precedent for the structural type.

Root + suffix	Surface	Derivational category	Surface meaning	Developed in
<i>*de- + *-r-</i>	<i>*de-r-</i>	athematic consonant-stem	locative body-part noun (δάρις, <i>derma</i>)	Paper 1, §3
<i>*de- + *-u_ṛ-/-w(e/o)n-</i>	<i>*deh₃-w_ṛ-</i>	heteroclite	gift-noun cluster (δῶρον, <i>dōnum</i> , <i>*durno-</i>)	Paper 1, §4–5
<i>*de- + *-r- + *-o-</i>	<i>*(t)ero-</i>	thematic adjectival	bilateral-contrastive suffix	Paper 2
<i>*de- + *-u_ṛ- + *-o-/-i-</i>	<i>*du-o-/du-i-</i>	thematic / adverbial	numeral “two” / “in two”	Paper 2 v10
<i>*de- + *-h₃-</i>	<i>*deh₃-</i>	extended verbal stem	“give” (post-Anatolian)	Paper 3
<i>*de- + postposed *-a</i>	<i>*de + a → *dā-</i>	univerbation, directive	“take” (Anatolian)	Paper 4

2.3 What the present paper does and does not claim

Paper 1’s specific contribution is the lexical branch: the two top rows of the skeleton table. The body-part noun **de-r-* and the heteroclitic gift-noun **de- + *-u_ṛ-/-w(e/o)n-* are the two derivational outputs that produce the comparative cluster Pokorny assembled at IEW 203. The framework does not propose a new PIE root. The bare **de-* and the locative **-r-* are independently motivated in Paper 3. The heteroclitic paradigm and the **w_ṛ > *ru* metathesis are Clayton’s published reconstructions, not framework innovations.

3. The lexical branch I: the body-part noun **de-r-*

3.1 Hesychian δάρις: the primary Greek witness

Hesychian δάρις/δάρις, glossed as “span of all fingers” (Attic equivalent σπῆλαι), is the direct Greek-internal surface witness for **de-r-*. Under the framework, this is the i-stem derivative **de-r-i-s* with regular Greek treatment: a zero-grade or e-grade formation on the compositional body-part noun, meaning “the span of the hand-direction.” The form is morphologically transparent and requires no special phonological commitments beyond the framework’s core claim that **de-r-* is a real compositional body-part noun.

3.2 Old Irish *derna* and Continental Celtic **DERNV*

The unmarked feminine singulative **de-r-nā-* yields Old Irish *derna*, Gaelic *deàrna* “palm of the hand,” and Continental Celtic **DERNV* (Bronze of Novallas, Stifter 2022 §10). The same Novallas bronze further attests the allative preposition *DV* (Beltrán Lloris et al. 2021; Prósper 2022) — Celtiberian reflex of Paper 3’s bare **de-* directional particle — placing both branches of the bare hand-element in a single inscription.

3.3 The locative **-r-* foundation

Three independent published treatments converge on the locative-spatial status of **-r-*: Lipp (2009) reconstructs **-er* and **-én* as parallel locative endings of acrostatic heteroclite neuters; Bauhaus (2019) argues that **-r* is a true PIE locative case ending (“a real case ending, not an ‘Adverbialendung’”); Kahl (2024) traces the secondary comparative of Indo-Iranian and Greek to “an archaic locative morpheme **-er*.” The compositional **de-r-* is therefore “at the place of the hand-direction” — the open palm as the body’s anchor for spatial reference.

3.4 The relation to Paper 3’s verbal **deh₃-*

Paper 3 §5 decomposes the give-verb **deh₃-* as bare **de-* + a verbal extension **-h₃-*. The verbal **deh₃-* and the body-part **de-r-* are therefore sibling formations on the same bare element. The heteroclitic gift-noun of §4 is built on the verbal stem **deh₃-*, with Clayton’s heteroclite paradigm operating over the post-Anatolian extended form: **de-r-* on the bare element at the pre-PIE stratum; **deh₃-wr̥-* on the extended verbal stem at the post-Anatolian PIE stratum.

4. The lexical branch II: the heteroclitic gift-noun

4.1 Clayton’s **-wr̥-/w(e/o)n-* heteroclite class

Clayton’s 2023 dissertation (UCLA) reconstructs the PIE **-wr̥-/w(e/o)n-* heteroclite class systematically and argues that the metathesis **wr̥ > *ru*

operated at Proto-Nuclear-Indo-European depth — post-Anatolian, pre-branch-separation. Crucially, Clayton independently identifies **deh₃-* as a root that forms heteroclites of this class, with examples Sanskrit *dārú-* “liberal” (from **déh₃-w_ṛ-*) and Prasun *pyṅrū* “gift” (from **pro-deh₃-w_ṛ-*). This is independent of the present project.

4.2 δῶπον as the metathesized strong stem

Under Clayton’s framework: (1) PNIE **deh₃-w_ṛ-* (heteroclitic strong stem); (2) PNIE metathesis **w_ṛ > *ru*, giving **deh₃-ru-*; (3) laryngeal **h₃* colors and lengthens the preceding vowel, yielding **dō-ru-*; (4) thematicization gives δῶπ-ο-ν; (5) Greek surface form δῶπον. The *Paradebeispiel* for the metathesis is **k^wetw_ṛ-* “four-” → **k^w(e)tru-* in Avestan *caθru-*, Gaulish *petru-*, Latin *quadru-* (Clayton 2023: §2.2; cf. Lubotsky 1994: 98). Further examples: **smók-w_ṛ-* “beard” → **smók-ru-* (Sanskrit *śmáśru-*); **swek-w_ṛ-h₂-* “mother-in-law” → (Sanskrit *śvaśrū-*, OCS *svekry*).

4.3 δάνος and Latin *dōnum*: the heteroclitic weak stem

If δῶπον is the metathesized strong stem, then δάνος “gift, loan” is the heteroclitic weak stem **deh₃-w(e/o)n-* with regular zero-grade vocalization. Latin *dōnum* is a thematicized derivative of the weak stem. Stifter’s (2022: §8) registration of “a heteroclitic r/n-stem as a recognized morphological possibility for the **deh₃-* verbal family” is now specifically Clayton’s reconstructed paradigm. The cognate set — δῶπον (strong, metathesized), δάνος (weak, zero-grade), Latin *dōnum* (weak, full grade, thematized), Sanskrit *dārú-* (strong, metathesized; Clayton), Celtic **durno-* (strong, metathesis + revocalization, §5), Latvian *dūfe* (strong, Baltic development; Schrijver) — is coherent under a single heteroclitic paradigm.

4.4 The “palm-breadth” sense of δῶπον

Greek δῶπον is attested with two meanings from the earliest Homeric layer: “gift” and “palm-breadth” (a unit of measurement equal to four finger-widths, one-quarter of a foot), attested in *Iliad* 4.109, Hesiod’s *Works and Days*, Nikander’s *Theriaca* 398, Vitruvius’s *De architectura* 2.3.3, and Pollux’s *Onomasticon* 2.157. Under the framework, the body-part “palm-breadth” sense is preserved by family-association with the locative-extended body-part noun **de-r-* on the same bare element. Vitruvius supplies the most explicit ancient testimony: “δῶπον Graeci appellant palmum, quod munus datum graece δῶπον appellatur, id autem semper geritur per manus palmam” (*De architectura* 2.3.3). Beekes’s two-lemma solution misses this family relationship.

5. Celtic **durno-* and the metathesis pathway

5.1 The cognate set: *u*-vocalism is original in Celtic

Celtic **durno-* “fist, hand” is attested with consistent *u*-vocalism in all four major branches: Old Irish *dorn*, Welsh *dwrn*, Breton *dorn/dourn*, Cornish *dorn*, and — critically — Continental Celtic Gaulish *Durnomagus*, *Durnacos*, *Dagodurnus* in first-century-BCE inscriptions. The Gaulish evidence is decisive: the *u* was already present in Continental Celtic before any Brythonic-internal sound change could have applied. As Schrijver (1995: 58) notes, this rules out the Morris Jones (1913: 87) raising rule **o > *u / __rn* as the source of the Brythonic vocalism. Ball & Müller (2009: 121) independently confirm that the relevant raising is Welsh-internal, not proto-Celtic.

5.2 The chain from PNIE **deh₃-wr̥-* to Celtic **durno-*

The compound stem **durno-* is the Celtic reflex of the heteroclitic strong stem of **deh₃-*, after PNIE metathesis. The chain: (1) PNIE **deh₃-wr̥-*; (2) PNIE metathesis **wr̥ > *ru* (Clayton 2023): **deh₃-ru-*; (3) revocalization to **ur-* in non-final position (cf. PIIr. **druna-* > OKhotanese *durna-*; Bailey 1961; Clayton 2023); (4) compounding with singulative **-no-*: **dur-no-*; (5) loss of the laryngeal in pre-Celtic. The OKhotanese parallel is the closest phonological analog for the Celtic development available anywhere in the literature.

5.3 The Lubotsky mechanism: σύρξ and πέσσυρες

Lubotsky (1994: 99–100) independently documents the phonological mechanism for the **-ru-* > **-ur-* revocalization: Greek σύρξ plausibly reflecting **twurk-* < **twr̥k-* — a non-final-syllable case where a syllabic resonant adjacent to a labial **w* is u-colored to surface as **ur*. Lesbian πέσσυρες < **k^wetwurV-* < **k^wetwrV-* is given as direct support. Lubotsky (1997) confirms the cross-linguistic naturalness of the phonological pressure via the Indo-Iranian pretonic **CRHuV > CuRvV / CruuV* rule.

5.4 Old Irish *derna* and **durno-* as derivational siblings

The Celtic evidence preserves both formations on the same root in the same branch: *derna* “palm” (the locative-extended body-part noun) alongside **durno-* “fist, hand” (the heteroclitic strong stem with metathesis). Two surface stems that the standard handbooks have not previously linked are derivational siblings surfacing from two different derivational pathways at two different chronological strata. The body-part meaning is confirmed by compound formations: Proto-Celtic **are-durnon* (Old Irish *aurdorn*, Welsh *arddwrn*, Breton *arzorn*) means “wrist”; Proto-Celtic **ambi-durnos* (Old Irish *imdorn*) means “sword hilt”. Neither compound makes sense with “fist” as the base element.

6. The Breton synchronic witness

6.1 The body-part dual *daouarn*

Ball & Müller (2009: 445–446) document that Breton *dorn* “hand” participates in the body-part dual system: the dual form is *daouarn* “hands” (from *daou* “two” + *dorn*), formed on exactly the same template as all other paired body parts — *lagad/daoulagad* “eyes,” *askell/divaskell* “wings,” *bronn/divronn* “breasts,” *glin/daoulin* “knees.” The dual category applies exclusively to anatomically paired body parts. Breton *dorn*’s membership confirms that its base meaning is the hand as a bilateral anatomical unit, not “fist” as a posture.

6.2 The *-iad* suffix and *dornad* “handful”

Breton *dorn* → *dornad* “handful” (Ball & Müller 2009: 804) employs the suffix marking the content a container holds. A “fistful” is incoherent as a content measure; a “handful” — the open palm as natural measure — is the normal formation. The derivation presupposes an open-hand reading of the base noun, exactly what Vitruvius’s $\delta\omega\rho\upsilon\nu$ = *palmus* identification (§4.4) confirms for Greek.

6.3 Breton as synchronic surface case

The Breton dual *daouarn* combines *daou* “two” with *dorn* transparently. Under the framework’s broader architecture (§2.2; Paper 2 v10), the inherited Celtic **durno-* itself contains a **-u-* element from the heteroclitic **-wṛ̥-/w(e/o)n-* paradigm. Breton is doing openly in *daouarn* what the inherited lexicon does opaquely in **durno-*: marking the paired body part with a pairedness morpheme. For Paper 1’s purposes, the synchronic Breton fact confirms the inherited Celtic **durno-* names the hand as a paired body part.

7. Engaging Beekes: the *u*-vocalism objection

7.1 Beekes’s two-lemma split of $\delta\omega\rho\upsilon\nu$

Beekes’s *Etymological Dictionary of Greek* (EDG s.v. $\delta\omega\rho\upsilon\nu$) treats $\delta\omega\rho\upsilon\nu$ as having two unrelated lemmas (gift-sense vs. palm-breadth-sense) and notes that Celtic and Baltic forms “point to PIE **u*, which excludes any relationship with $\delta\omega\rho\upsilon\nu$.” The framework revises both positions. Against the two-lemma split: Vitruvius’s explicit etymological pairing of the two senses (§4.4) is the ancient Greek-internal evidence that the gift-sense and the palm-breadth-sense are derivational siblings. Against the *u*-vocalism objection: the Celtic *u* is not inherited PIE root vocalism but the PNIE-stage metathesized reflex of **-wṛ̥-* revocalized as **-ur-* (the Lubotsky $\sigma\upsilon\rho\xi$ -type mechanism). Once the **u* is recognized as this metathesized reflex, Beekes’s objection dissolves.

7.2 Albanian *dorë* and the Demiraj 1990 removal

The framework does not contest Demiraj’s (1990) derivation of Albanian *dorë* “hand” from PIE **ǵhesr-* rather than from the **durno-* family. The framework’s argument runs without Albanian: the comparative cluster Greek δῶρον + Hesychian δάρις + Celtic **durno-/*derna-* + Latvian *dūfe* is sufficient. PIE has at least two body-part words for “hand:” **ǵhesr-* (Greek χεῖρ, Hittite *keššar*, Albanian *dorë*) and the framework’s **de-r-/durno-* family.

7.3 δ- might reflect **du-*: Beekes/Schrijver and the seed of Paper 2

Schrijver (1995: 58), citing Beekes, observes that “in the Greek forms, δ- might reflect **du-* (cf. δωδεκά ‘12’ < **duō-*).” The framework’s architecture (§2.2) takes this observation seriously. Paper 2 v10 develops it using Dunkel (2014, s.v. **uī*), who reconstructs an inherited compound **uī-tero-* attested as Vedic *vítarām* “further, sideways,” Avestan *vítara-*, and Old Church Slavonic *vŭtoru* “the second” — a transparent **uī-* + **-tero-* compound whose meaning is the bilateral ordinal “the second one of two,” and whose **uī-* Dunkel traces to dissimilation from compound-first-member **duī* “two.” Beekes’s δ- ← **du-* observation is the seed; Paper 2 v10 grows the argument.

8. The Baltic and Italic evidence

8.1 Latvian *dūfe*

Latvian *dūfe*, *dūris* “fist” enters the cognate set as a third Indo-European witness to the heteroclitic strong stem. Schrijver (1995: 58) lists it explicitly among the “possible cognates” of Welsh *dwrn* and analyzes the long *ū* as reflecting earlier **uH*. Derksen (2015) takes an agnostic stance. The framework registers Latvian *dūfe* as supporting evidence but not load-bearing; the Greek and Celtic evidence is sufficient to anchor the lexical claim.

8.2 Latin *dōnum* and *dare*

Latin *dōnum* “gift” parallels Greek δάνοϛ as the Italic reflex of the heteroclitic weak stem **deh₃-w(o)n-* — thematicized, with the long *ō* from laryngeal-coloring of the full grade. The body-part senses preserved in Greek and Celtic have no Italic surface witness, but Italic preserves both the verbal stem (Latin *dare*) and the heteroclitic gift-noun (*dōnum*).

9. The Anatolian convergence

The framework receives independent semantic support from Anatolian, on grounds developed by Norbruis (2021: 243–249) without reference to the present project. Norbruis argues, on Anatolian-internal morphological evidence, that the original meaning of **deh₃-* was “to take,” not “to give”: the

give-meaning is a post-Anatolian innovation. His semantic development: “to take” > (with Goal or Beneficiary constituent) “to bring, convey, deliver” > (when animate) “to hand, to give.” Paper 4 develops this insight into a full formal account of Hittite *dā-*. Anatolian’s absence from Clayton’s metathesis sample is exactly what the layered chronology predicts: the heteroclitic **-wr-/w(e/o)n-* paradigm, including its PNIE metathesis, operates on the extended form **deh₃-* and is therefore post-Anatolian.

10. Forward references: Papers 2, 3, 4

10.1 Paper 2: the contrastive suffix **-tero-* and the numeral **duo-/dui-*

Paper 2 (Chavis & Claude 2026b) reconstructs the contrastive suffix **-tero-* compositionally as **de- + *-r- + *-o-* with suffix-initial devoicing of **d-* to **t-*, and documents the bilateral-contrastive function across the IE family. Paper 2 v10 additionally develops the numeral branch **duo-/dui-* “two” as a fourth derivational outcome, drawing on Dunkel (2014)’s published etymology that **u₁-* “arose by dissimilation from the compound first member **dui* ‘two’”.

10.2 Paper 3: the directional particle **de/do*, the verbal **deh₃-*, and the locative **-r-*

Paper 3 (Chavis & Claude 2026c) develops three foundational pieces: the directional particle **de/do* “to, toward” reconstructed as the bare **de-* in its grammaticalized adverbial function, attested as Greek allative *-δε* (productive in Mycenaean by 1400 BCE), Old Church Slavonic *do*, English *to*, and Celtiberian *DV* in the Bronze of Novallas; the verbal **deh₃-* decomposed as bare **de-* + verbal extension **-h₃-*; and the locative **-r-* grounded in three independent published treatments (Lipp 2009, Bauhaus 2019, Kahl 2024).

10.3 Paper 4: the Anatolian directive **de + a* → Hittite *dā-*

Paper 4 (Chavis & Claude 2026d) derives Hittite *dā-* “to take” from a pre-Anatolian univerbation **de + a*, where bare **de-* combines with a postposed directional **a* “toward (the speaker),” contracting by hiatus to long **ā*. This is the sixth row of the skeleton table, paralleling Bauhaus’s (2019) documented locative/directive paradigmatic pairs on other PIE roots.

11. Engaging the wider literature

Pokorny 1959 (IEW 203). Pokorny grouped together the comparative cluster without committing to a specific reconstruction. The framework’s contribution is to provide the formal etymology Pokorny did not work out.

Schrijver 1995. Schrijver lists δωρ- / δαρ- / Celtic **durno-* / Latvian *dūfe* as “possible cognates” and describes the formal relation as “obscure.” His observation that the Celtic *u* is original is one the framework accepts. The framework supplies the explicit formal relation Schrijver flagged as obscure: heteroclitic strong stem (PNIE metathesis) for δωρ-, **durno-*, *dūfe*; locative-extended body-part noun for δαρ- and Celtic **derna-*.

Matasović 2009 (s.v. **durno-*). Matasović classifies the Celtic word as a non-IE loanword after the removal of Albanian *dorë*. The framework offers exactly the alternative IE etymology that was missing: the Celtic *u* is PNIE **-wṛ-* metathesized and revocalized, with the morphological framework supplied by Clayton (2023). Matasović’s substrate hypothesis is no longer the best available account.

Mayrhofer 1992 (EWAia, s.v. *ādara-*). Mayrhofer rejects Grassmann’s (1875) **der-* with semantics “esteem, respect, regard” — a different semantic field entirely. The framework proposes **de-r-* “hand” on a different evidence base: Greek δάρις, Celtic **durno-/*derna-*, Latvian *dūfe*.

Clayton 2023. Clayton does not make the Celtic **durno-* or the Greek δῶρον connections; he is studying the heteroclitic class systematically. Those connections are the contribution of the present paper. The PIr. **druna-* > OKhotanese *durna-* parallel that Clayton cites (from Bailey 1961) is the closest phonological analog for Celtic **durno-*.

Lubotsky 1994 and 1997. Lubotsky 1994 establishes the σύρξ / πέσσυρες pattern of u-coloring of syllabic resonants in labial environments. Lubotsky 1997 establishes the Indo-Iranian pretonic **CRHuV > CuRvV / CruuV* rule. Neither paper is making claims about the Celtic word; both supply the phonological mechanisms the framework’s derivation requires.

12. Costs, alternatives, and what we do not claim

The reconstruction of the body-part noun as compositional **de-r-* is non-standard. It does not propose a new PIE root — the bare **de-* and the locative **-r-* are independently motivated in Paper 3 — but it does propose a body-part noun that has not been in the standard handbooks. Reviewers committed to atomistic root reconstruction will resist the compositional analysis on principle.

The heteroclitic analysis of δῶρον / **durno-* is similarly non-standard, though the underlying framework (Clayton 2023) is itself standard scholarship. The paper combines two pieces that the literature has not previously combined: Clayton’s **-wṛ-/-(e/o)n-* heteroclitic class and the bare-element architecture.

The PNIE metathesis pathway for Celtic **durno-* requires more steps than older reconstructions positing Celtic-internal *u*-development. The compensating advantage is that the same derivation explains the relation to δῶρον, *dārú-*, and the rest of the cognate set in a way the older reconstructions could not.

Latvian *dūfe* is registered as supporting but not load-bearing. We do not claim **deh₃-* is wrong as the verbal pre-form for the give-verb. We do not claim that all body-part “hand” words across IE descend from **de-r-*: PIE **ǵh₂esr-* (Greek χεῖρ, Albanian *dorë*, Hittite *keššar*) is a separate hand-word. We do not claim Clayton’s analysis is the only possible reading of δῶρον; the older **deh₃-ro-* derivation remains a live alternative.

13. Probability assessment

The probability for Paper 1 alone — that the comparative cluster reflects two derivational siblings on the same bare hand-element **de-* — is assessed at approximately 30–40%.

Three factors support this range. *First, the Clayton 2023 framework supplies a morphological account that was previously lacking.* The §4 analysis is grounded in independent dissertation-level work on the **-wr₂-/w(e/o)n-* heteroclite class. *Second, the OKhotanese durna- parallel (Bailey 1961, cited in Clayton 2023) is the closest phonological analog for Celtic *durno- available anywhere in the literature.* Earlier treatments proposed multiple Celtic-internal candidates; the metathesis pathway replaces them with a single derivation with a documented cross-linguistic parallel. *Third, the unification of δωρ-/δαρ- under a single morphological architecture explains Schrijver’s “obscurity” rather than describing it.*

The marginal probability for the whole four-paper framework: 30–40% (Paper 1) × 35–45% (Paper 2) × 32–42% (Paper 3) × 30–42% (Paper 4) ≈ 1.0–3.2%. We continue to regard the central claim as more likely wrong than right, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

14. Conclusion

The Greek–Celtic–Baltic hand cluster has stood since the nineteenth century as a comparative set whose formal derivation was unresolved. Pokorny (IEW 203) listed it; Schrijver (1995: 58) called the formal relation “obscure”; Matasović (2009) classified the Celtic word as substrate; Beekes (EDG s.v. δῶρον) split δῶρον into two unrelated lemmas and rejected the Celtic/Baltic connection on grounds of *u*-vocalism. The present paper offers an Indo-European etymology under the polyfunctional architecture of the broader four-paper project,

supplemented by Clayton’s (2023) reconstruction of the **-wr̥-/w(e/o)n-* heteroclit class.

The body-part noun proper is the compositional **de-r-* (= bare **de-* of Paper 3 + locative **-r-*, supported by Lipp 2009, Bauhaus 2019, Kahl 2024). It is preserved in Hesychian δάρις (Greek zero-grade i-stem body-part form), Old Irish *derna* (unmarked feminine singulative **de-r-nā-*), and Continental Celtic **DERNV* (Bronze of Novallas, Stifter 2022).

The gift-noun family — Greek δῶρον, δάνοϛ, Latin *dōnum*, Sanskrit *dārú-*, Prasun *pyṅrū*, Celtic **durno-*, Latvian *dūfe* — is the **-wr̥-/w(e/o)n-* heteroclit of **deh₃-* in Clayton’s reconstruction. Beekes’s *u*-vocalism objection is dissolved by the metathesis architecture. The same Novallas bronze that preserves **DERNV* further attests the allative preposition *DV* (Beltrán Lloris et al. 2021; Prósper 2022) — Celtiberian reflex of Paper 3’s bare **de-* directional particle — placing both branches of the bare hand-element in a single inscription.

15. Authorship and contributions

Chavis’s contributions. The central hypothesis originates with Chavis. The identification of the dual attestation of Greek δῶρον — gift and palm-breadth — as evidence of a single etymological family was the project’s founding insight. Chavis directed the research agenda at every stage: articulating the hypothesis that **-tero-* descends from the same root (“the *ter* in **-tero-* is hand”); identifying *ambidexter* as the morphologically transparent compound witness for Paper 2; posing the question “whence comes the *r* in **-tero-*?” that opened the compositional analysis of Paper 3; identifying the contestability of Melchert’s χαμαί-as-allative reading that created analytical space for the framework’s alternative; hypothesizing that Hittite *dā-* “to take” preserves a pre-Anatolian **de-* + postposed directional **a* construction; identifying PIE postpositional syntax as dissolving the central morphosyntactic objection to Paper 4’s claim; and directing the incorporation of Clayton (2023) as the central new structural piece that resolved the Celtic *u* puzzle. The decision to make Paper 1 the project’s morphological anchor — restructuring it around the polyfunctional architecture of §2 as centerpiece, with the δῶρον heteroclitic analysis, the Celtic **durno-* metathesis pathway, the Breton synchronic witness, and the substantive Beekes engagement as the spine of the lexical argument — is Chavis’s. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who also supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and analytical development. Across the project’s arc, Claude located and read the relevant published literature — including Lipp

(2009), Bauhaus (2019), and Kahl (2024) as three independent published treatments converging on the locative **-r-*; Clayton (2023) on the **-wr-/w(e/o)n-* heteroclit class and the PNIE metathesis; Norbruis (2021) on the semantic chronology of **deh₃-*; Stifter (2022) on the Bronze of Novallas and the **DERNV* attestation; Lubotsky (1994, 1997) on the σύρξ-type phonological mechanism and the pretonic vocalization rule; Bailey (1961, via Clayton) on the OKhotanese *durna-* parallel; Dunkel (2014) on the **ui-tero-* compound and the **ui- ← *dui-* dissimilation etymology; Kloekhorst (2008, 2013) on the Hittite paradigm and the TA/DA phonetic distinction; Hoffner & Melchert (2008) on the synchronic Hittite preverb and adverb systems; Yakubovich (2008) on the CAPERE hieroglyph and its acrophonic derivation; Byrd (2015) on PIE syllable structure and the OCP mechanism, including footnote 161 identifying suffix-initial deletion as an explicit alternative; Lejeune (1961) and van Hes (2021) on Mycenaean -δε; Boley (2007) as historiographic precursor; Lehmann (1974) and Friedrich (1975) on PIE postpositional syntax; Ball & Müller (2009) on Breton body-part duals; and ancillary sources throughout. Claude drafted the prose of all four papers from Chavis’s structural and substantive direction; proposed the Anatolian distribution argument (Paper 2 §8) and developed the distributional test across Sanskrit, Avestan, Latin, and Greek; surfaced the typological parallels (Uralic, Tungusic, Mayan, P’urhepecha) for the cross-cultural sections; identified Bauhaus’s locative-directive paradigmatic pairs as published structural precedent for Papers 3–4; and integrated the architectural alignment across all four papers at each revision stage. The probability estimates in all four papers are Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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